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CREATE DATABASE StdCoachingSystem;

CREATE TABLE Student (
    StudentID INT IDENTITY(1,1) PRIMARY KEY,
    Name NVARCHAR(255),
    AcademicYear INT
);

CREATE TABLE Coach (
    CoachID INT IDENTITY(1,1) PRIMARY KEY,
    Name NVARCHAR(255),
    Specialization NVARCHAR(255)
);

CREATE TABLE FavoriteCoach (
    FavoriteID INT IDENTITY(1,1) PRIMARY KEY,
    CoachID INT,
    StudentID INT,
    FOREIGN KEY (CoachID) REFERENCES Coach(CoachID),
    FOREIGN KEY (StudentID) REFERENCES Student(StudentID)
);

CREATE TABLE Session (
    SessionID INT IDENTITY(1,1) PRIMARY KEY,
    CoachID INT,
    StudentID INT,
    SessionDate DATETIME,
    FOREIGN KEY (CoachID) REFERENCES Coach(CoachID),
    FOREIGN KEY (StudentID) REFERENCES Student(StudentID)
);

ALTER TABLE Session
ADD SessionStatus NVARCHAR(50);

UPDATE Session
SET SessionStatus = 'Completed'
WHERE SessionID= 1;

UPDATE Session
SET SessionStatus = 'Canceled'
WHERE SessionID= 2;

CREATE TABLE Package (
    PackageID INT IDENTITY(1,1) PRIMARY KEY,
    StudentID INT,
    Description NVARCHAR(255),
    Price DECIMAL(10, 2),
    SessionCount INT,
    FOREIGN KEY (StudentID) REFERENCES Student(StudentID)
);

CREATE TABLE Assignment (
    AssignmentID INT IDENTITY(1,1) PRIMARY KEY,
    StudentID INT,
    Description NVARCHAR(255),
    DueDate DATETIME,
    Status NVARCHAR(50),
    FOREIGN KEY (StudentID) REFERENCES Student(StudentID)
);

CREATE TABLE Feedback (
    FeedbackID INT IDENTITY(1,1) PRIMARY KEY,
    StudentID INT,
    CoachID INT,
    SessionID INT,
    Comments NVARCHAR(255),
    Rating INT,
    FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
    FOREIGN KEY (CoachID) REFERENCES Coach(CoachID),
    FOREIGN KEY (SessionID) REFERENCES Session(SessionID)
);

CREATE TABLE Schedule (
    ScheduleID INT IDENTITY(1,1) PRIMARY KEY,
    CoachID INT,
    AvailableDate DATETIME,
    FOREIGN KEY (CoachID) REFERENCES Coach(CoachID)
);

CREATE TABLE Availability (
    AvailabilityID INT IDENTITY(1,1) PRIMARY KEY,
    CoachID INT,

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    Day NVARCHAR(50),
    StartTime TIME,
    EndTime TIME,
    FOREIGN KEY (CoachID) REFERENCES Coach(CoachID)
);

CREATE TABLE Goal (
    GoalID INT IDENTITY(1,1) PRIMARY KEY,
    StudentID INT,
    Description NVARCHAR(255),
    TargetDate DATE,
    ProgressStatus NVARCHAR(255),
    FOREIGN KEY (StudentID) REFERENCES Student(StudentID)
);

CREATE TABLE ResourceMaterial (
    ResourceID INT IDENTITY(1,1) PRIMARY KEY,
    Title NVARCHAR(255),
    Description NVARCHAR(255),
    FileURL NVARCHAR(255),
    UploadedBy INT,
    PackageID INT,
    Timestamp DATETIME,
    FOREIGN KEY (UploadedBy) REFERENCES Coach(CoachID),
    FOREIGN KEY (PackageID) REFERENCES Package(PackageID)
);

CREATE TABLE ActionPlan (
    ActionPlanID INT IDENTITY(1,1) PRIMARY KEY,
    CoachID INT,
    StudentID INT,
    GoalID INT,
    TaskDescription NVARCHAR(255),
    DueDate DATETIME,
    FOREIGN KEY (CoachID) REFERENCES Coach(CoachID),
    FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
    FOREIGN KEY (GoalID) REFERENCES Goal(GoalID)
);

CREATE TABLE ProgressReport (
    ReportID INT IDENTITY(1,1) PRIMARY KEY,
    StudentID INT,
    Description NVARCHAR(255),
    ReportDate DATETIME,
    FOREIGN KEY (StudentID) REFERENCES Student(StudentID)
);

CREATE TABLE Certificate (
    CertificateID INT IDENTITY(1,1) PRIMARY KEY,
    ProgressReportID INT,
    StudentID INT,
    PackageID INT,
    Title NVARCHAR(255),
    DateAwarded DATE,
    FOREIGN KEY (ProgressReportID) REFERENCES ProgressReport(ReportID),
    FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
    FOREIGN KEY (PackageID) REFERENCES Package(PackageID)
);

CREATE TABLE RewardSystem (
    RewardID INT IDENTITY(1,1) PRIMARY KEY,
    StudentID INT,
    AssignmentID INT,
    RewardType NVARCHAR(255),
    Points INT,
    DateEarned DATE,
    FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
    FOREIGN KEY (AssignmentID) REFERENCES Assignment(AssignmentID)
);

CREATE TABLE CommunicationParticipant (
    ParticipantID INT IDENTITY(1,1) PRIMARY KEY,
    Role NVARCHAR(255),
    UserID INT,
    UserType NVARCHAR(255)
);

CREATE TABLE CommunicationLog (
    CommunicationID INT IDENTITY(1,1) PRIMARY KEY,
    SenderID INT,
    RecipientID INT,
    MessageContent NVARCHAR(255),

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        Timestamp DATETIME,
        FOREIGN KEY (SenderID) REFERENCES CommunicationParticipant(ParticipantID),
        FOREIGN KEY (RecipientID) REFERENCES CommunicationParticipant(ParticipantID)
    );

CREATE TABLE Notification (
    NotificationID INT IDENTITY(1,1) PRIMARY KEY,
    ParticipantID INT,
    NotificationType NVARCHAR(255),
    Content NVARCHAR(255),
    RelatedEntityID INT,
    RelatedEntityType NVARCHAR(255),
    IsRead BIT,
    FOREIGN KEY (ParticipantID) REFERENCES CommunicationParticipant(ParticipantID)
);

CREATE TABLE Payment (
    PaymentID INT IDENTITY(1,1) PRIMARY KEY,
    StudentID INT FOREIGN KEY REFERENCES Student(StudentID),
    AmountPaid DECIMAL(10, 2),
    PaymentDate DATETIME,
    PaymentMethod NVARCHAR(50),
    TransactionID NVARCHAR(100),
    Status NVARCHAR(50)
);

CREATE TABLE Attendance (
    AttendanceID INT IDENTITY(1,1) PRIMARY KEY,
    SessionID INT FOREIGN KEY REFERENCES Session(SessionID),
    StudentID INT FOREIGN KEY REFERENCES Student(StudentID),
    Status NVARCHAR(50),
    Timestamp DATETIME
);

CREATE TABLE Enrollment (
    EnrollmentID INT IDENTITY(1,1) PRIMARY KEY,
    StudentID INT FOREIGN KEY REFERENCES Student(StudentID),
    CourseID INT FOREIGN KEY REFERENCES Course(CourseID),
    EnrollmentDate DATETIME,
    Status NVARCHAR(50)
);

CREATE TABLE Course (
    CourseID INT IDENTITY(1,1) PRIMARY KEY,
    Title NVARCHAR(255) NOT NULL,
    Description NVARCHAR(500),
    Duration NVARCHAR,
    Price DECIMAL
);

-- Insert statements
INSERT INTO Student (Name, AcademicYear)
VALUES
('Emir', 6),
('Elif', 8);

INSERT INTO Coach (Name, Specialization)
VALUES
('Ege', 'Mathematics'),
('Gamze', 'Science');

INSERT INTO Package (StudentID, Description, Price, SessionCount)
VALUES
(1, 'Basic Math Package', 100.00, 5),
(2, 'Science Exploration Package', 150.00, 8);

INSERT INTO Session (CoachID, StudentID, SessionDate)
VALUES
(1, 1, '2025-01-03 10:00:00'),
(2, 2, '2025-01-04 11:00:00');

INSERT INTO ResourceMaterial (Title, Description, FileURL, UploadedBy, PackageID, Timestamp)
VALUES
('Math Basics', 'A basic guide to mathematics', 'http://example.com/math.pdf', 1, 1, '2025-01-01 10:00:00'),
('Science Primer', 'Introduction to science concepts', 'http://example.com/science.pdf', 2, 2, '2025-01-02 11:00:00');

INSERT INTO Assignment (StudentID, Description, DueDate, Status)
VALUES
(1, 'Complete Math Worksheet', '2025-01-10', 'Pending'),
(2, 'Write Science Essay', '2025-01-15', 'Completed');

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INSERT INTO RewardSystem (StudentID, AssignmentID, RewardType, Points, DateEarned)
VALUES
(1, 1, 'Badge', 50, '2025-01-10'),
(2, 2, 'Certificate', 100, '2025-01-15');

INSERT INTO Feedback (StudentID, CoachID, SessionID, Comments, Rating)
VALUES
(1, 1, 1, 'Great session on algebra!', 5),
(2, 2, 2, 'Excellent explanation of physics!', 4);

INSERT INTO FavoriteCoach (CoachID, StudentID)
VALUES
(1, 1),
(2, 2);

INSERT INTO Goal (StudentID, Description, TargetDate, ProgressStatus)
VALUES
(1, 'Master Algebra', '2025-02-01', 'In Progress'),
(2, 'Understand Newton Laws', '2025-03-01', 'Not Started');

INSERT INTO ProgressReport (StudentID, Description, ReportDate)
VALUES
(1, 'Mid-term math progress', '2025-01-15'),
(2, 'Science project evaluation', '2025-01-20');

INSERT INTO Availability (CoachID, Day, StartTime, EndTime)
VALUES
(1, 'Monday', '09:00:00', '12:00:00'),
(2, 'Tuesday', '13:00:00', '16:00:00');

INSERT INTO Schedule (CoachID, ScheduledDates)
VALUES
(1, '2025-01-03 10:00:00'),
(2, '2025-01-04 11:00:00'),
(1, '2025-01-05 15:00:00'),
(2, '2025-01-06 14:00:00');

INSERT INTO ActionPlan (CoachID, StudentID, GoalID, TaskDescription, DueDate)
VALUES
(1, 1, 1, 'Complete algebra exercises', '2025-01-15'),
(2, 2, 2, 'Prepare science experiment report', '2025-01-20');

INSERT INTO CommunicationParticipant (Role, UserID, UserType)
VALUES
('Student', 1, 'Student'),
('Coach', 2, 'Coach');

INSERT INTO CommunicationLog (SenderID, RecipientID, MessageContent, Timestamp)
VALUES
(1, 2, 'Can we schedule a session for algebra?', '2025-01-02 09:00:00'),
(2, 1, 'Sure, do it on Monday.', '2025-01-02 09:30:00');

INSERT INTO Notification (ParticipantID, NotificationType, Content, RelatedEntityID, RelatedEntityType, IsRead)
VALUES
(1, 'Reminder', 'Your session is scheduled for 2025-01-03.', 1, 'Session', 0),
(2, 'Update', 'You have received feedback on your session.', 2, 'Feedback', 1);

INSERT INTO Payment (StudentID, AmountPaid, PaymentDate, PaymentMethod, TransactionID, Status)
VALUES
(1, 150.00, '2025-01-05 14:30:00', 'Credit Card', 'TX123456789', 'Completed'),
(2, 200.00, '2025-01-06 10:00:00', 'PayPal', 'TX987654321', 'Pending');

INSERT INTO Attendance (SessionID, StudentID, Status, Timestamp)
VALUES
(1, 1, 'Present', '2025-01-03 10:00:00'),
(2, 2, 'Absent', '2025-01-04 11:00:00');

INSERT INTO Enrollment (StudentID, CourseID, EnrollmentDate, Status)
VALUES
(1, 1, '2025-01-01 09:00:00', 'Active'),
(2, 2, '2025-01-02 09:30:00', 'Completed');

INSERT INTO Course (Title, Description, Duration, Price)
VALUES
('Introduction to Mathematics', 'Basic concepts in mathematics.', 4, 120.00),
('Advanced Science', 'In-depth study of science topics.', 8, 200.00);

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-- Additional SQL Operations
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-- 1. CREATE
-- 2. ALTER
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alter table schedule
drop column availabledate
alter table schedule
add ScheduledDates datetime

-- 3. INSERT
INSERT INTO Student ( Name, AcademicYear)
VALUES ('Tuna', 4);

INSERT INTO Session (CoachID, StudentID, SessionDate)
VALUES
(1, 1, '2025-01-12 10:00:00');

-- 4. UPDATE
UPDATE Goal
SET ProgressStatus = 'Completed'
WHERE GoalID = 1 AND StudentID = 2;

-- 5. DELETE
DELETE FROM Session
WHERE SessionID = 3;

-- 6. SELECT with GROUP BY
SELECT PackageID, COUNT(StudentID) AS TotalStudents
FROM Package
GROUP BY PackageID;

-- 7. SELECT with ORDER BY
SELECT StudentID, FeedbackID, Rating
FROM Feedback
ORDER BY Rating DESC;

-- 8. SELECT with WHERE
SELECT Comments
FROM Feedback
WHERE StudentID = 2 AND CoachID = 2;

-- 9. SELECT with INNER JOIN
SELECT s.StudentID, s.Name AS StudentName, c.Name AS CoachName, se.SessionID
FROM Student s
INNER JOIN Session se ON s.StudentID = se.StudentID
INNER JOIN Coach c ON se.CoachID = c.CoachID;

-- 10. SELECT with WHERE, GROUP BY, and ORDER BY
SELECT StudentID, AVG(Rating) AS AvgRating
FROM Feedback
WHERE Rating > 0
GROUP BY StudentID
ORDER BY AvgRating DESC;

-- Trigger procedure and view

--PROCEDURE
GO
CREATE PROCEDURE AddGoal
    @student_id INT,
    @description NVARCHAR(255),
    @start_date DATE,
    @end_date DATE
AS
BEGIN
    INSERT INTO Goal (StudentID, Description, TargetDate, ProgressStatus)
    VALUES (@student_id, @description, @end_date, 'Pending');
END;

--Usage
EXEC AddGoal
    @student_id = 1,
    @description = 'Complete Algebra Basics',
    @start_date = '2025-01-06',
    @end_date = '2025-01-20';

-- TRIGGER
GO
CREATE TRIGGER NotifyStudentOnFeedback
ON Feedback
AFTER INSERT
AS
BEGIN
    SET NOCOUNT ON;

```

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INSERT INTO Notification (ParticipantID, NotificationType, Content, RelatedEntityID, RelatedEntityType, IsRead)
SELECT
    i.StudentID,
    'Feedback',
    CONCAT('New feedback received: ', i.Comments),
    i.FeedbackID,
    'Feedback',
    0
FROM
    INSERTED i;
END;

--Usage
INSERT INTO Feedback (StudentID, CoachID, SessionID, Comments, Rating)
VALUES (1, 1, 1, 'Great session!', 5);

SELECT * FROM Notification;
select * from Feedback;

INSERT into Feedback (StudentID, CoachID, SessionID, Comments, Rating)
values (2, 2, 1, 'You should check your spellings', 3) ;

--VIEW
GO
CREATE VIEW StudentProgressView AS
SELECT
    s.StudentID,
    s.Name AS StudentName,
    pr.ReportID AS ProgressID,
    pr.Description AS ActivityName,
    f.Rating AS Score,
    pr.ReportDate AS CompletionDate
FROM
    Student s
JOIN
    ProgressReport pr ON s.StudentID = pr.StudentID
LEFT JOIN
    Feedback f ON s.StudentID = f.StudentID AND pr.ReportID = f.SessionID;

GO

--Usage
select * from StudentProgressView

GO
CREATE VIEW CommunicationSummary AS
SELECT
    cl.CommunicationID,
    cl.MessageContent,
    cl.Timestamp,
    Sender.ParticipantID AS SenderID,
    Sender.Role AS SenderRole,
    Sender.UserType AS SenderUserType,
    Recipient.ParticipantID AS RecipientID,
    Recipient.Role AS RecipientRole,
    Recipient.UserType AS RecipientUserType
FROM
    CommunicationLog cl
JOIN
    CommunicationParticipant Sender
    ON cl.SenderID = Sender.ParticipantID
JOIN
    CommunicationParticipant Recipient
    ON cl.RecipientID = Recipient.ParticipantID;
GO

SELECT *
FROM CommunicationSummary;

SELECT *
FROM CommunicationSummary
WHERE SenderRole = 'Coach';

```